

Solution to the uniqueness of Dunford decomposition

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1 Problem Statement

Let $n \in \mathbb{N}$, and let $N, N', S, S' \in M_n(\mathbb{R})$ such that:

- S and N commute ($SN = NS$),
- S' and N' commute ($S'N' = N'S'$),
- S and S' are diagonalizable,
- N and N' are nilpotent,
- $S + N = S' + N'$.

Prove that $S = S'$ and $N = N'$.

Rather than the usual proof, we will solve this problem considering $P(S+N)$ for a well-chosen polynomial $P \in \mathbb{R}[X]$. The polynomial is obtained by Hermite interpolation: we prescribe not only its values at the eigenvalues of S , but also its first $n - 1$ derivatives there.

I stumbled across this problem as a nineteen-year-old and decided to give it a go, finding this solution by myself. At the time, I had no knowledge of the Dunford decomposition. I am aware that this problem has been solved for a long time, but I wanted to share the proof I found.

2 Notation

Since S is diagonalizable in $M_n(\mathbb{R})$, its eigenvalues are real. We write $\lambda_1, \dots, \lambda_r$ for its *distinct* eigenvalues, and $d_1, \dots, d_n$ for its eigenvalues repeated according to multiplicity, so that the d_j are exactly the diagonal entries of a diagonalization of S and $\{d_1, \dots, d_n\} = \{\lambda_1, \dots, \lambda_r\}$ as sets. Note also that $N^n = 0$, since a nilpotent matrix of $M_n(\mathbb{R})$ has nilpotency index at most n .

3 Solution

3.1 Step 1: Spectral properties of S and S'

The matrices S and N commute, and both have a characteristic polynomial that splits over $\mathbb{R}$: that of S because S is diagonalizable in $M_n(\mathbb{R})$, and that of N because N is nilpotent, so $\chi_N = X^n$. Two commuting matrices with split characteristic polynomials are simultaneously trigonalizable, so there is a basis in which S and N are both upper triangular.

In such a basis, the diagonal of S carries its eigenvalues $d_1, \dots, d_n$, repeated according to multiplicity. Since N is nilpotent, in such a basis, N is strictly upper triangular. Consequently $S+N$ is upper triangular with the same diagonal as S , and therefore:

S and $S+N$ have the same eigenvalues, with the same multiplicities. The same argument gives that S' and $S' + N'$ have the same eigenvalues, with the same multiplicities. Since $S+N = S' + N'$, we conclude that S and S' have the same eigenvalues, with the same multiplicities.

3.2 Step 2: Polynomial representation

Let $P \in \mathbb{R}[X]$ be a polynomial:

$$P(X) = \sum_{k=0}^d a_k X^k.$$

Since S is diagonalizable, it can be written as:

$$S = Q \begin{pmatrix} d_1 & & \\ & \ddots & \\ & & d_n \end{pmatrix} Q^{-1},$$

where $d_1, \dots, d_n$ are the eigenvalues of S repeated according to multiplicity. Using the commutativity of S and N , we apply the binomial theorem to $P(S+N)$:

$$P(S+N) = \sum_{k=0}^d a_k \sum_{i=0}^k \binom{k}{i} Q \begin{pmatrix} d_1^{k-i} & & \\ & \ddots & \\ & & d_n^{k-i} \end{pmatrix} Q^{-1} N^i.$$

Rearranging the sums gives:

$$P(S+N) = \sum_{i=0}^d \frac{1}{i!} Q \begin{pmatrix} \sum_{k=i}^d a_k \frac{k!}{(k-i)!} d_1^{k-i} & & \\ & \ddots & \\ & & \sum_{k=i}^d a_k \frac{k!}{(k-i)!} d_n^{k-i} \end{pmatrix} Q^{-1} N^i,$$

which simplifies to:

$$P(S + N) = \sum_{i=0}^d \frac{1}{i!} Q \begin{pmatrix} P^{(i)}(d_1) & & \\ & \ddots & \\ & & P^{(i)}(d_n) \end{pmatrix} Q^{-1} N^i.$$

3.3 Step 3: Choosing the polynomial P

Suppose P is chosen such that:

- $P(\lambda_j) = \lambda_j$ for all $j \in \llbracket 1, r \rrbracket$,
- $P^{(i)}(\lambda_j) = 0$ for all $j \in \llbracket 1, r \rrbracket$ and all $i \in \llbracket 1, n-1 \rrbracket$.

Since every d_m is one of the λ_j , these conditions give $P(d_m) = d_m$ and $P^{(i)}(d_m) = 0$ for $1 \leq i \leq n-1$ and every $m \in \llbracket 1, n \rrbracket$.

In the expansion of Step 2, the term $i = 0$ is then $Q \operatorname{diag}(d_1, \dots, d_n) Q^{-1} = S$; the terms with $1 \leq i \leq n-1$ vanish because the diagonal matrix is zero; and the terms with $i \geq n$ vanish because $N^i = 0$. Hence

$$P(S + N) = S.$$

The same polynomial works for S' , since S and S' have the same eigenvalues by Step 1, so $P(S' + N') = S'$. As $S + N = S' + N'$, it follows that $S = S'$.

3.4 Step 4: Existence of the polynomial P

The conditions of Step 3 prescribe, at each of the r distinct points $\lambda_1, \dots, \lambda_r$, the value of P and its first $n-1$ derivatives. This is exactly a Hermite interpolation problem, and it always has a solution.

Consider the linear map

$$\Phi : \mathbb{R}_{rn-1}[X] \longrightarrow \mathbb{R}^{rn}, \quad P \longmapsto \left(P(\lambda_j), P'(\lambda_j), \dots, P^{(n-1)}(\lambda_j) \right)_{1 \leq j \leq r},$$

where $\mathbb{R}_{rn-1}[X]$ denotes the polynomials of degree strictly less than rn . Both spaces have dimension rn , so it suffices to show that Φ is injective.

Let $P \in \ker \Phi$. For each j , the first n derivatives of P at λ_j (orders 0 to $n-1$) vanish, so λ_j is a root of P of multiplicity at least n . The λ_j being pairwise distinct, the polynomial $\prod_{j=1}^r (X - \lambda_j)^n$, of degree rn , divides P . Since $\deg P < rn$, this forces $P = 0$.

Hence Φ is bijective, and there exists a polynomial P of degree less than rn satisfying all the conditions of Step 3.

3.5 Step 5: Conclusion

Applying $P(S + N) = S$ and $P(S' + N') = S'$, and given $S + N = S' + N'$, we obtain $S = S'$. Subtracting from $S + N = S' + N'$ then gives $N = N'$.